Thus, $a = \dfrac{1}{2}$ and so, $\log_x 2 = \dfrac{1}{2}$ which gives $x^{\frac{1}{2}} = 2$.
Thus, $x = 2^2 = 4$.

Example 2.38 Solve $x^{\log_3 x} = 9$.

Solution:
Let $\log_3 x = y$.
Then $x = 3^y$ and so, $3^{y^2} = 9$.
Thus, $y^2 = 2$, which implies $y = \sqrt{2}, \quad -\sqrt{2}$. Hence, $x = 3^{\sqrt{2}}, \ \ 3^{-\sqrt{2}}$.

Example 2.39 Compute $\log_3 5 \log_{25} 27$.

Solution:
$\log_3 5 \log_{25} 27 = \log_3 5 \log_{25} 3^3$.
$= \log_3 5 \times 3 \log_{25} 3$ (by exponent rule)
$= 3 \log_{25} 5 = \frac{3}{\log_5 25} = \frac{3}{2 \log_5 5} = \frac{3}{2}$.

Example 2.40 Given that $\log_{10} 2 = 0.30103$, $\log_{10} 3 = 0.47712$ (approximately), find the number of digits in $2^8.3^{12}$.

Solution:
Suppose that $N = 2^8 3^{12}$ has $n + 1$ digits. Then N can be written as $10^n \times b$ where $1 \leq b < 10$.
Taking logarithm to the base 10, we get
$\log N = \log(10^n\ b) = n \log 10 + \log b = n + \log b$.
On the other hand,
$\log N = \log 2^8 3^{12} = 8 \log 2 + 12 \log 3 = 8 \times 0.30103 + 12 \times 0.47712 = 8.13368$.
Thus, we get $n + \log b = 8.13368$. Since $1 \leq b < 10$ the number of digits is 9.

Exercise - 2.12

1. Let $b > 0$ and $b \neq 1$. Express $y = b^x$ in logarithmic form. Also state the domain and range of the logarithmic function.
2. Compute $\log_9 27 - \log_{27} 9$.
3. Solve $\log_8 x + \log_4 x + \log_2 x = 11$.
4. Solve $\log_4 2^{8x} = 2^{\log_2 8}$.
5. If $a^2 + b^2 = 7ab$, show that $\log \dfrac{a+b}{3} = \dfrac{1}{2}(\log a + \log b)$.
6. Prove $\log \dfrac{a^2}{bc} + \log \dfrac{b^2}{ca} + \log \dfrac{c^2}{ab} = 0$.
7. Prove that $\log 2 + 16 \log \dfrac{16}{15} + 12 \log \dfrac{25}{24} + 7 \log \dfrac{81}{80} = 1$.
8. Prove $\log_{a^2} a \ \log_{b^2} b \ \log_{c^2} c = \frac{1}{8}$.
9. Prove $\log a + \log a^2 + \log a^3 + \cdots + \log a^n = \dfrac{n(n+1)}{2} \log a$.